



MICROSOFT INNOVATION CLUB, VIT CHENNAI
Adaptive Climate Impact Optimizer for Digital, Built, and
Natural Systems
Sustainability & Climate

1. BACKGROUND

Digital infrastructure, buildings, campuses, cities, and natural ecosystems consume or influence energy, water, carbon, biodiversity, and environmental quality. Decisions in these systems are usually optimized in isolation — AI requests are routed for speed or cost without considering regional carbon intensity, buildings use electricity despite available daylight, and environmental monitoring identifies pollution or ecosystem stress without selecting the most effective intervention.

Organizations often have historical records, sensor readings, satellite data, climate feeds, or operational logs, but use them mainly for *dashboards and reports rather than decisions*.

The Problem

- Environmental cost is invisible at decision time — carbon, water, ecological, and pollution impacts are calculated after resources have already been consumed
- Data does not become intervention — monitoring tools identify conditions but rarely select or implement the next practical action
- Objectives conflict — lower emissions may increase latency or cost, while ecological improvements may compete for land, water, maintenance, or budget
- Models operate with sparse observations — ocean, water-quality, biodiversity, energy, and resource data may have missing locations or time periods

- Projected benefits are not verified — a recommendation is presented without comparing prediction against measured outcome

Target beneficiaries: campuses, facility teams, cloud and AI platforms, municipalities, researchers, environmental organizations, and communities, with planners, sustainability officers, utilities, regulators, and affected citizens as secondary beneficiaries.

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Sustainability & Climate — Problem Statement

